PUKPARKING

Functional Specification

North-West University (NWU) Potchefstroom Campus

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Functional Specifications Document

Authorization Memorandum

I have carefully assessed the Functional Specifications Document for the **PukParking System**.

MANAGEMENT CERTIFICATION - Please check the appropriate statement.

_____ The document is accepted.

_____ The document is accepted pending the changes noted.

_____ The document is not accepted.

We fully accept the changes as needed improvements and authorize initiation of work to proceed. Based on our authority and judgment, the continued development and implementation of this system is authorized.

NAME	SIGNATURE	DATE
_____ **Project Manager**	_____	_____
_____ **Sponsor**	_____	_____
_____ **Director**	_____	_____

Table of Contents

1. Introduction 4
1.1 Purpose 4
1.2 Reference Documents 4
1.3 Abbreviations and Acronyms 4
2. General Description 5
2.1.1 Product Context 5
2.1.2 System Objectives 5
2.1.3 Scope 6
2.2 User Classes and Characteristics 7
2.3 Overview of Functional Requirements 8
2.3.1 Public Access and Destination Search 8
2.3.2 Interactive Map and Parking Information 8
2.3.3 Parking Eligibility, Filtering and Recommendation 8
2.3.4 Walking Navigation and Alternative Routing 8
2.3.5 Parking Full Reporting and Crowd-Assisted Availability 9
2.3.6 Administrator Management 9
2.3.7 Progressive Web App and Offline Access 9
2.4 Overview of Data Requirements 9
2.4.1 Entity Relational Diagram 11
2.4.2 Entity Description 12
2.5 Operating Environment 14
2.6 General Constraints, Assumptions, Dependencies, Guidelines 14
2.6.1 Constraints 14
2.6.2 Assumptions 15
2.6.3 Dependencies 15
2.6.4 Guidelines 15
2.7 User Documentation 15
3.1 External Interface Requirements 15
3.1.1 User Interface 15
3.1.2 Software Interface 15
3.1.3 Communications Interface 16
3.2 Functional Requirements 16
3.3 Non-Functional Requirements 22
3.4 Use Case 25
3.5 Appendix. Issues List 26

1. Introduction

1.1 Purpose

This Functional Specification (FS) defines what PukParking must do for students, staff, visitors and administrators at the North-West University (NWU) Potchefstroom Campus. It establishes the functional boundary of the system and describes the actors that interact with it, the inputs they provide, the processing and decisions the system must perform, the outputs it must present, the information it must maintain, and the acceptance basis used during testing and User Acceptance Testing (UAT). The Functional Specification focuses on required behaviour rather than implementation technology. The accompanying Technical Specification (TS) defines how the approved requirements will be implemented. This distinction keeps the functional requirements traceable to the business problem and allows the completed system to be evaluated against an agreed requirements baseline.

1.2 Reference Documents

The following project documents form the approved requirements baseline for this Functional Specification:

- PukParking Business Case (2026) - defines the business problem, business opportunity, alternative solutions, recommended solution and implementation approach.
- PukParking Feasibility Study (2026) - defines the business drivers, business requirements, feasibility assessment and the basis for selecting PukParking.
- PukParking Project Overview (2026) - defines the project objectives, user and administrator functionality, parking recommendation and reassignment logic, PWA behaviour and the original application scope.

1.3 Abbreviations and Acronyms

Acronym	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DSRM	Design Science Research Methodology
ETA	Estimated Time of Arrival
FR	Functional Requirement
FS	Functional Specification
GPS	Global Positioning System
HTTPS	Hypertext Transfer Protocol Secure
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
NWU	North-West University
POPIA	Protection of Personal Information Act 4 of 2013
PWA	Progressive Web Application
REST	Representational State Transfer
TS	Technical Specification
UAT	User Acceptance Testing
UID	Unique Identifier
UI/UX	User Interface / User Experience

2. General Description

2.1.1 Product Context

PukParking is a mobile-first Progressive Web Application designed for the NWU Potchefstroom Campus. Its purpose is to help a campus user identify a suitable parking area for a selected destination instead of driving around campus without reliable guidance. A user can search for a building, identify the parking category that applies to them, review parking information and receive a recommendation for the nearest valid parking area. The recommendation considers walking distance, parking restrictions, whether the parking area is active and current crowd-assisted availability. The system then provides a walking route from the recommended parking area to the selected destination.

The system is required because the current parking process depends heavily on physical parking discs, signage, static campus maps and users' prior knowledge of the campus. The Business Case and Feasibility Study identify parking-search time, unnecessary vehicle circulation, parking-rule compliance, visitor accessibility, accurate parking information and NWU's move toward digital services as the main business drivers for the project.

PukParking was selected as the preferred solution after comparison with an existing parking application/platform and a sensor-based smart parking system. The Feasibility Study ranked PukParking first with a weighted score of 8.85/10, while the Business Case rated it 54/60. It was selected because it provides the strongest alignment with NWU-specific requirements, can be customised for campus parking rules, can be introduced without installing physical parking sensors, supports future expansion, and avoids dependence on a third-party parking product while retaining external mapping services where required.

2.1.2 System Objectives

The objectives of PukParking are to:

- Recommend the closest valid parking area for a selected campus destination.
- Reduce parking-search time and unnecessary vehicle circulation on campus
- Represent NWU parking types and restrictions digitally so that users are directed only to suitable parking areas.
- Provide accurate walking directions, distance and estimated walking time from a selected or recommended parking area to the destination building.
- Use crowd-assisted "Parking Full" reporting to reduce repeated recommendations of parking areas that are likely to be full.
- Provide an interactive campus map showing buildings, parking areas and relevant parking information.
- Allow authorized administrators to maintain building and parking information without changing application code.
- Provide an installable and responsive PWA that works across phones, tablets and desktop browsers and retains approved cached campus information during connectivity loss.
- Deliver the MVP without public user accounts or physical parking sensors, keeping the first implementation simple and accessible.

2.1.3 Scope

In Scope

- Login-free public access for students, staff and visitors.
- Building search by full name or abbreviation and direct destination selection from the map.
- An interactive campus map showing buildings, parking areas and the user's current location when browser location permission is granted.
- Parking-category filtering and display of parking type, capacity, estimated occupancy, restrictions, operating hours, accessible-parking availability and an image where available.
- Recommendation of the nearest valid parking area using walking distance, parking restrictions, active status and crowd-assisted occupancy information.
- Walking-route generation from the selected parking area to the destination, including distance and estimated walking time.
- A "Parking Full" action that records temporary crowd-assisted availability information, lowers the affected area's recommendation priority, recalculates an alternative and expires the temporary report after a configurable short period of approximately 10-15 minutes.
- Authenticated administrator functions to add, edit, delete and enable/disable buildings and parking areas and to maintain restrictions, capacity, operating hours, accessibility and parking images.
- Installable PWA behaviour, responsive layouts and offline access to previously cached application resources and approved campus information.

Out of scope for the MVP

- End-user registration, user profiles or persistent end-user location history.
- Physical parking-bay sensors, cameras or automated occupancy-detection hardware.
- Parking-space reservations.
- Push or email notifications to public users.
- Native Android or iOS applications.
- Automated integration with NWU vehicle access-control or boom-gate systems.
- Guaranteed live Mapbox routing, geocoding or crowd-report submission while the device is offline.

2.2 User Classes and Characteristics

PukParking has two access levels: public campus users and authenticated administrators. The public interface is shared by students, staff and visitors and does not require an account. Their parking permissions differ, so the system must allow the applicable parking category to be selected or applied when filtering recommendations. Administrators use a protected management interface to maintain the campus data used by the public application.

User Class	Role/Access	Expected volume	Technical Characteristics
Student	Public user; no login. Find parking permitted for the applicable student/accessible category, search destinations, review parking details, navigate and report a parking area as full.	High and variable; potentially many users during peak campus periods.	Mixed. Expected to be familiar with basic mobile/browser use, but the interface must remain simple for less confident users.
Staff	Public user; no login. Find parking permitted for the applicable staff/accessible category, search destinations, navigate and report full parking areas.	Medium to high, depending on campus activity.	Mixed to medium. Expected to be familiar with browser-based university services.
Visitor	Public user; no login. Find visitor/accessible parking and navigate without prior knowledge of the campus, building abbreviations or parking-zone system.	Low to medium but may increase during events and registration periods.	Unknown/mixed. The design must support first-time and low confidence users with clear language and minimal steps.
Administrator	Authenticated user. Maintain buildings and parking records, restrictions, capacities, operating hours, accessibility, images and active status; keep public information accurate.	Low; limited to authorized administrative staff.	Medium to high after administrator training; expected to use data management forms and validation messages

2.3 Overview of Functional Requirements

The following groups summarize the main functions that PukParking must provide. Detailed, prioritised and testable Functional Requirement (FR) statements are specified later in this document.

2.3.1 Public Access and Destination Search

- Students, staff and visitors must be able to use the public application without registering or signing in.
- Users must be able to search for a campus building by full name or abbreviation, select a building from the results and view available building information.
- A building may also be selected directly from the interactive map.

2.3.2 Interactive Map and Parking Information

- The system must display campus buildings and parking areas on an interactive map with pan and zoom controls.
- The user's current location should be shown when browser geolocation permission is granted; denying permission must not prevent destination search or parking recommendation.
- Each parking area must present its name, parking type, capacity, estimated occupancy, restrictions, operating hours, accessible-parking availability and image where available.

2.3.3 Parking Eligibility, Filtering and Recommendation

- Users must be able to select the parking category applicable to them, including student, staff, visitor, accessible and open parking where applicable, before parking recommendations are filtered.
- The system must exclude parking areas that are inactive or incompatible with the selected parking category or restrictions.
- For a selected destination, the recommendation process must consider walking distance, parking restrictions, active status and current crowd-assisted occupancy information and return the best valid parking area.
- If no valid parking area is available, the system must clearly inform the user instead of recommending an invalid area.

2.3.4 Walking Navigation and Alternative Routing

- The system must generate a walking route from the selected or recommended parking area to the destination building.
- The route must display walking distance and estimated walking time.
- If another valid parking area is selected or recommended, the route must be recalculated for that parking area.

2.3.5 Parking Full Reporting and Crowd-Assisted Availability

- The current recommendation must provide a “Parking Full” action when the user finds that parking area full.
- Submitting the report must trigger immediate recalculation and return the next suitable valid parking area when one is available.
- Each active report must temporarily increase the confidence that the parking area is full; repeated reports within a short period must further reduce that area’s recommendation priority.
- Temporary reports must expire automatically after a configurable period of approximately 10-15 minutes so that the parking area can return to normal recommendation priority.
- The reporting process must remain anonymous and must not require a public user account.

2.3.6 Administrator Management

- Only authenticated administrators may access the management portal.
- Administrators must be able to add, edit and delete building and parking-area records.
- Administrators must be able to maintain parking type, restrictions, capacity, operating hours, accessible-parking information, parking images and active/inactive status.
- Changes approved by an administrator must be reflected in the public data used for search, display and recommendation.

2.3.7 Progressive Web App and Offline Access

- PukParking must be installable as a PWA on supported devices and remain responsive on phones, tablets and desktop browsers.
- The PWA must cache application resources and approved campus information so that previously loaded information can still be accessed during connectivity loss.
- Functions that require live external services or server updates, including fresh Mapbox routing and new Parking Full reports, require connectivity; the application must clearly indicate when live data cannot be refreshed.

2.4 Overview of Data Requirements

As described in the PukParking description document, this web application will not require a user to create an account to use the application, therefore this application only requires more permanent data to function. This includes information about each Building on campus, each Parking Area where students/staff and visitors can park, details of the administrators that will be able to edit the created Buildings and Parking Areas.

The minimum information needed to search and display a destination building is:

- Building ID (system-generated).
- Administrator ID (Foreign Key linking to the registering administrator).
- Name.
- Abbreviation.
- Description.
- Coordinates (Latitude and Longitude).

The minimum information required to store a parking record is:

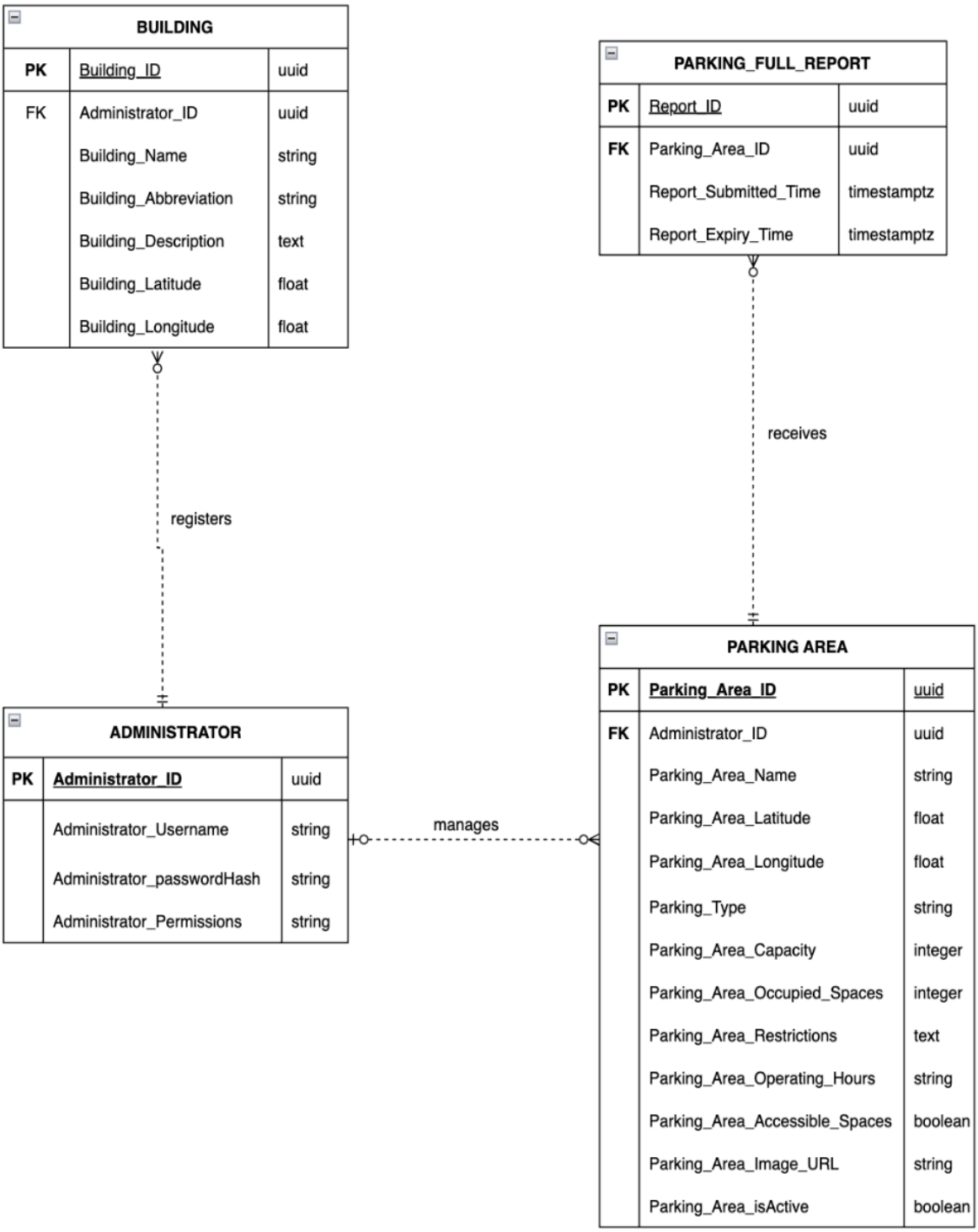
- Parking Area ID (system-generated).
- Administrator ID (Foreign Key linking to the registering administrator).
- Name.
- Coordinates.
- Parking type (e.g., staff, student, visitor).
- Total capacity.
- Occupied spaces.
- Operational status.
- Access restrictions.
- Operating hours.
- Accessible parking availability.
- Image URL (Optional).

The minimum information required to store and create an administrator profile:

- Admin ID (system-generated).
- Username.
- Hashed password.
- Permission levels to grant management access.

Also, to support the crowd-sourced parking reassignment feature as described in the project description document, full reports are created with timestamp and expiry, from which the system derives a temporary occupancy confidence score for each parking area.

2.4.1 Entity Relational Diagram



2.4.2 Entity Description

BUILDING:

Represents a campus destination that a user can search for or select on the map. Buildings are read-only for end users and are only ever written to through the administration portal.

Field	Data Type	Description
Building_ID	uuid	Primary Key. System-generated identifier used to reference a building.
Administrator_ID	uuid	Foreign Key linking to the ADMINISTRATOR who registered the building.
Building_Name	string	Full display name of the campus building.
Building_Abbreviation	string	Short code used for fast searching (e.g., "E3").
Building_Description	text	Details about the building, faculties, or departments inside etc.
Building_Latitude	float	Geographic latitude coordinate used for Haversine distance calculations.
Building_Longitude	float	Geographic longitude coordinate used for Haversine distance calculations.

PARKING_AREA:

Represents a physical parking area on campus. This is the entity the recommendation and reassignment algorithms operate on, therefore it carries the fields those algorithms need restrictions, active status, and a baseline capacity figure, in addition to the descriptive fields shown to users.

Field	Data Type	Description
Parking_Area_ID	uuid	Primary Key. System-generated identifier used to reference a parking area.
Administrator_ID	uuid	Foreign Key linking to the ADMINISTRATOR who created or last updated the lot details.
Parking_Area_Name	string	Display name of the parking lot (e.g., "Engineering North Parking").

Parking_Area_Latitude	float	Geographic latitude coordinate for route generation via the Mapbox API.
Parking_Area_Longitude	float	Geographic longitude coordinate for route generation via the Mapbox API.
Parking_Type	string	Designated user group of the given parking area (Students, Staff or Visitor).
Parking_Area_Capacity	integer	Total number of parking bays in the parking area.
Parking_Area_Occupied_ Spaces	integer	Number of currently estimated or recorded occupied parking bays in the parking area.
Parking_Area_Restrictions	text	Plain text or encoded permit restrictions enforced for the parking area.
Parking_Area_Operating_ Hours	string	Active hours during which restrictions or gate access apply.
Parking_Area_Accessible_ Spaces	boolean	Boolean flag indicating if disabled/handicapped parking bays are available.
Parking_Area_Image_URL	string	Path or file string to an uploaded photo of the parking area.
Parking_Area_isActive	boolean	Boolean value which indicates whether a parking area is currently enabled.

ADMINISTRATOR:

The only entity requiring authentication. Represents a staff member with access to the administration portal used to manage buildings and parking areas. Kept separate from the public-facing entities because ordinary users never authenticate, so administrator credentials should not sit on the same table as content that anonymous users read.

Field	Data Type	Description
Administrator_ID	uuid	Primary Key. Unique administrative user ID.
Administrator_Username	string	Login handle used for administrative portal authentication.
Administrator_passwordHash	string	The bcrypt hash of the administrator's account password.
Administrator_Permissions	string	Permission flags that describes what the administrator account is allowed to do.

PARKING_FULL_REPORT:

Backs the crowd-sourced occupancy-confidence mechanism described under the Parking Reassignment Algorithm. Each row is one "Parking Full" button press, not a running total, which is used to let the individual reports expire independently after their configurable time window instead of a single counter having to be reset. A parking area's current occupancy confidence is derived at request time by counting its still-unexpired reports, rather than being stored as a mutable field.

Field	Data Type	Description
Report_ID	uuid	Primary Key. Unique report entry identifier.
Parking_Area_ID	uuid	Foreign Key linking to the PARKING_AREA which the report was raised against.
Report_Submitted_Time	timestamptz	Timestamp recording when the report was submitted.
Report_Expiry_Time	timestamptz	Expiration timestamp after which the unavailability penalty lifts.

2.5 Operating environment

The system is web-based and will be built using a compilation of different technologies. The frontend will be built with React and TypeScript and deployed using Vite, with Tailwind CSS handling responsive styling through utility classes. The system is configured as a progressive web app using the Vite PWA plugin to allow the app to be installed to a home screen.
The backend runs on Node.js with Express.js providing the REST API layer. The database of choice is MongoDB and administrator authentication will be handled by JWT statelessly.
The interactive map of the campus is rendered client-side using Mapbox GL JS. The Mapbox Directions API generates turn-by-turn walking routes between the user and a selected parking area, while the Mapbox Geocoding API converts text-based location searches into coordinates. To avoid third party dependencies, the user's current position is determined using the native browser Geolocation API.

Client and public visitors may use any modern device with a browser.

2.6 General constraints, Assumptions, dependencies, guidelines

2.6.1 Constraints

- The system must be operational by 16 October 2026 to align with the module schedule. Testing windows are unavailable during NWU exam and semester-test periods.
- Hardware will not be procured as mentioned in the Business Case. The system is mainly web based and does not need any additional hardware for it to operate.
- The administrator can work from any NWU supplied PC; no additional servers need to be purchased.
- Network coverage, some areas may have weak Wi-Fi connectivity, and the system may not work optimally.

- NWU IT policy - what's allowed on campus network, data governance rules, approval processes
- Campus safety policy

2.6.2 Assumptions

- NWU IT will provide server and hosting capacity for the backend of the system
- NWU facilities/security will take ownership of admin access post-deployment
- Users won't need multi-language support and can use the system in English
- Internet is available for 90% of the time all over campus

2.6.3 Dependencies

- Connectivity: reliable Wi-Fi for the user to access the system on the browser
- A modern browser with no specific browser API dependency.
- Manual input from users to update parking availability

2.6.4 Guidelines

These are the guidelines that matter the most

- Minimize personal data collection. The system does not need to identify the user unless necessary
- Zero-onboarding principle to ensure minimal manual intervention is required to make it easier for the user.
- Design the entities to support expansion without making any schema changes

2.7 User Documentation

The following user documentation will be delivered:

- A manual for the administrative staff covering login, maintaining of buildings, maintaining of parking areas and how to set the system up.
- A developer manual for the future expansion of the project

3.1 External interface requirements

3.1.1 User interface

The wireframes for this system are included in the Technical Specifications document. The wireframes are subject to change but so far, we have the Home page, About us page, Admin login, Maintenance, user landing page, user navigation(interactive map), and parking directions

3.1.2 Software interface

The system depends on the following software services

- Mapbox GL JS
- Mapbox Directions API

- Mapbox Geocoding API
- Browser Geolocation API
- MongoDB

3.1.3 Communications interface

- HTTPS only (TLS 1.2+) for all browser-to-server traffic between the React frontend and the Express API; HTTP requests redirected to HTTPS; HSTS enabled with a one-year max-age.
- Frontend-to-backend communication conducted via Axios over REST, with CORS explicitly restricted to the deployed frontend origin only.
- Administrator authentication conducted via JWT bearer tokens attached to authenticated requests; no server-side session storage.
- MongoDB connection secured in transit (TLS), with the database not exposed to the public internet - accessible only from the backend's hosting environment.
- Mapbox GL JS, Directions API, and Geocoding API accessed over HTTPS from the client, authenticated via a scoped Mapbox API key.

3.2 Functional Requirements

FR1: Access Parking Information

The system shall provide public access to PukParking parking information without requiring users to register or sign in.

- Students, staff and guests shall be able to access the public application without creating an account.
- The system shall allow users to select their User Category: Staff, Student or Guest.
- The selected User Category shall be used when determining which parking areas are available to the user.
- The system shall provide access to the main parking-search functionality after the User Category has been selected.

FR2: Determine Current Location and Destination

The system shall allow the user to specify their current location and destination.

- The system shall automatically obtain and add the user's current location when browser geolocation permission is granted.
- The user shall be able to manually enter their current location if they do not want to use their current location or if geolocation is unavailable.
- The system shall allow the user to search for a campus building as their destination.
- The system shall allow the user to search for a building using its full name.
- The system shall allow the user to search for a building using its recognised abbreviation.
- The system shall allow the user to select a previously used destination from Building History.
- The system shall allow the user to proceed after a valid current location and destination have been specified.

FR3: Display Interactive Campus Map

The system shall display an interactive campus map after a destination has been entered or selected.

- The interactive map shall only be displayed after the user has entered or selected a destination.
- The map shall display the user's current or specified starting location.
- The map shall display the selected destination building.
- The map shall display a route from the current/specified location towards the destination building.
- The map shall display parking areas that are close to or located along the route to the destination.
- The map shall allow users to pan across the map.
- The map shall allow users to zoom in and out.
- The map shall provide sufficient visual distinction between buildings, routes and parking areas.
- The system shall update the displayed map when the current location or destination changes.

FR4: Display and Select Available Parking

The system shall display parking areas relevant to the user's journey and allow the user to select a suitable parking area.

- The system shall list the parking areas displayed on the map below the map.
- The system shall provide relevant parking options based on the user's selected User Category.
- The system shall display the name of each available parking area in the list.
- The system shall indicate when a parking area is full.
- A full parking area shall not display an active Select button.
- Users shall be able to select an available parking area from the list.
- The system shall allow the user to select a parking area directly from the map where applicable.
- When a parking area is selected, the system shall display a summary of the selected parking area's information in a dropdown or expandable section.
- The user shall be able to proceed to confirm the selected parking area.

FR5: Confirm Parking Selection

The system shall provide a confirmation page before the user proceeds to the selected parking area.

- The system shall open a new page after the user chooses to confirm a parking area.
- The confirmation page shall clearly identify the selected parking area.
- The system shall display detailed information about the selected parking area.
- The detailed information shall include the parking type.
- The detailed information shall include the parking capacity where applicable.
- The detailed information shall include current or estimated occupancy where available.
- The detailed information shall include applicable restrictions.
- The detailed information shall include operating hours.

- The detailed information shall include accessible-parking information where applicable.
- The detailed information shall include an approved image where available.
- The user shall be able to confirm the selected parking area.
- The user shall be able to return to the parking-selection screen if they do not wish to proceed.

FR6: Identify Parking Availability and Eligibility

The system shall determine whether parking areas are available and applicable to the selected User Category.

- The system shall identify active and inactive parking areas.
- The system shall identify parking areas that are available to Staff.
- The system shall identify parking areas that are available to Students.
- The system shall identify parking areas that are available to Guests.
- The system shall exclude parking areas that are incompatible with the selected User Category.
- The system shall exclude inactive parking areas from selection.
- The system shall apply applicable parking restrictions when determining whether a parking area can be selected.
- The system shall identify a parking area as full when its current availability indicates that it cannot accommodate additional vehicles.
- The system shall prevent a user from selecting a parking area that is identified as full.

FR7: Provide Parking Confirmation and Navigation

The system shall provide navigation to the confirmed parking area after the user confirms their selection.

- The system shall display an updated interactive map after parking confirmation.
- The map shall display the user's current or specified starting location.
- The map shall display the selected parking area.
- The map shall display a route from the current location to the selected parking area.
- The system shall update the route where the user's starting location changes and live location information is available.
- The system shall provide the route using the available mapping and routing service.
- The system shall inform the user when a route cannot be generated.

FR8: Display Parking Rules, Restrictions and Important Signs

The system shall provide the user with relevant rules and information after a parking area has been confirmed.

- The system shall display the rules applicable to the selected parking area.
- The system shall display parking restrictions applicable to the selected parking area.
- The system shall display important signs or instructions that the user should pay attention to.
- The system shall display operating restrictions where applicable.
- The system shall provide the information below the navigation map.
- The displayed rules and restrictions shall correspond to the selected parking area.

FR9: Record Parking Arrival and User Activity

The system shall allow the user to indicate their arrival at the selected parking area and record relevant parking activity.

- The user shall be able to indicate that they have arrived at the selected parking area.
- The system shall record the selected parking area as the user's current parking location.
- The system shall allow the user to indicate that the selected parking area is full upon arrival.
- The system shall allow the user to rate their parking experience.
- The system shall allow the user to provide a comment about their parking experience.
- The user shall be able to select Done after completing the parking activity.
- The system shall record the relevant activity and feedback for parking statistics.
- The system shall return the user to the home page after the parking activity has been successfully completed.

FR10: Report Parking as Full and Redirect User

The system shall allow users to report a parking area as full and provide an alternative parking option.

- The user shall be able to indicate that the selected parking area is full.
- The system shall record the Parking Full report against the relevant parking area.
- The Parking Full report shall be submitted anonymously.
- The system shall not require the user to create an account to submit a Parking Full report.
- A valid Parking Full report shall temporarily increase the confidence that the parking area is full.
- Multiple recent Parking Full reports shall further reduce the parking area's availability priority.
- The system shall automatically expire Parking Full reports after a configurable period of approximately 10–15 minutes.
- Expired reports shall no longer affect parking availability or selection.
- After a Parking Full report, the system shall identify another valid parking area where one is available.
- The alternative parking area shall comply with the user's selected User Category and applicable restrictions.
- The system shall display the alternative parking area as an available option.
- The system shall update the navigation route when the user selects the alternative parking area.
- If no alternative parking area is available, the system shall clearly inform the user.
- The Parking Full process shall be reflected in the parking statistics.

FR11: Return to Home After Successful Parking

The system shall return the user to the PukParking home page after successful completion of the parking process.

- The system shall allow the user to select Done after arriving and completing their parking activity.
- The system shall record the user's parking activity before returning to the home page.

- The system shall record applicable parking feedback before returning to the home page.
- The system shall return the user to the home page after successful submission.
- The user shall be able to begin a new parking search from the home page.

FR12: Secure Administrator Access

The system shall restrict access to the PukParking management portal to authenticated administrators.

- Administrators shall be required to authenticate before accessing the management portal.
- Unauthenticated users shall not be permitted to access administrator functionality.
- Public users shall not be able to modify building or parking information.
- Administrator permissions shall be enforced by the system.
- Administrator authentication credentials shall be securely handled.
- The system should record significant administrator activities for accountability.

FR13: Manage Building Information

The system shall allow authorised administrators to manage campus building information.

- Administrators shall be able to add building records.
- Administrators shall be able to edit building records.
- Administrators shall be able to delete building records.
- Administrators shall be able to maintain building names.
- Administrators shall be able to maintain building abbreviations.
- Administrators shall be able to maintain building location information.
- Administrators shall be able to maintain information required for destination search.
- Administrators shall be able to maintain information required for map display.
- Changes approved by an administrator shall be reflected in public building search and map functionality.
- Updated building information shall be available for future destination searches.

FR14: Manage Parking Area Information

The system shall allow authorised administrators to manage parking-area information.

- Administrators shall be able to add parking-area records.
- Administrators shall be able to edit parking-area records.
- Administrators shall be able to delete parking-area records.
- Administrators shall be able to maintain parking-area names.
- Administrators shall be able to maintain parking types.
- Administrators shall be able to maintain User Category eligibility.
- Administrators shall be able to maintain parking restrictions.
- Administrators shall be able to maintain parking capacity.
- Administrators shall be able to maintain occupancy information where applicable.
- Administrators shall be able to maintain operating hours.
- Administrators shall be able to maintain accessible-parking information.
- Administrators shall be able to maintain parking-area location information.

- Administrators shall be able to upload, replace and remove approved parking-area images.
- Administrators shall be able to maintain parking rules and important signs.
- Administrators shall be able to set a parking area as active or inactive.
- Approved changes shall be reflected in the public application's parking information and selection process.

FR15: Manage Parking Statistics

The system shall record and provide parking activity information for administrative statistics.

- The system shall record completed parking activities.
- The system shall associate recorded activity with the relevant parking area.
- The system shall record Parking Full reports.
- The system shall record applicable parking ratings.
- The system shall record user comments where submitted.
- The system shall record relevant activity timestamps.
- The system shall provide recorded information for the parking statistics page.
- The system shall distinguish between temporary Parking Full reports and completed parking activity.
- Administrators shall be able to view parking statistics through the management portal.

FR16: Provide Progressive Web App Functionality

The system shall operate as a Progressive Web App (PWA) on supported devices.

- The application shall be installable on supported devices.
- The application shall provide the required PWA manifest.
- The application shall use service-worker functionality where required.
- The application shall provide a responsive interface on mobile phones.
- The application shall provide a responsive interface on tablets.
- The application shall provide a responsive interface on desktop computers.
- Core interface components shall adapt to different screen sizes.

FR17: Provide Offline Access

The system shall provide access to previously loaded and approved information when connectivity is temporarily unavailable.

- The PWA shall cache required application resources.
- The PWA shall cache approved campus building and parking information that has previously been loaded.
- Users shall be able to view cached information during temporary connectivity loss.
- The system shall clearly indicate when information may be outdated.
- New Parking Full reports shall require connectivity.
- Fresh routing information shall require connectivity.
- The system shall clearly indicate when live information cannot be retrieved or refreshed.
- The system shall not present cached occupancy information as current live information.

3.3 Non-Functional Requirements

NFR1: Performance

The system shall provide acceptable response times during normal operating conditions.

- The initial application interface should load within approximately 3 seconds under normal network conditions.
- Building search results should be displayed within approximately 2 seconds after a valid search.
- The parking list should be displayed within approximately 3 seconds after the destination has been selected and the required data is available.
- A successful Parking Full report should trigger processing of the alternative parking options within approximately 5 seconds, excluding external-service delays.
- The interface shall remain responsive while maps and routes are loading.

NFR2: Usability

The system shall provide a simple and intuitive user experience.

- Users shall be able to understand the main parking process without formal training.
- The system shall clearly identify the current step in the parking process.
- Search, selection, confirmation, Parking Full and Done actions shall be clearly identifiable.
- The system shall provide clear feedback after user actions.
- Error messages shall be understandable to non-technical users.
- The system shall minimise unnecessary steps between selecting a destination and confirming parking.

NFR3: Accessibility

The system shall support accessibility for users with different abilities.

- Core functionality should support keyboard navigation.
- Text and controls shall provide sufficient contrast.
- The system shall not rely solely on colour to communicate important information.
- Images shall provide alternative text where appropriate.
- Interactive controls shall be sufficiently large for touch interaction.
- Important parking restrictions and warnings shall be presented in a clearly readable format.

NFR4: Responsiveness and Compatibility

The system shall operate effectively across supported devices and browsers.

- The application shall support modern desktop browsers.
- The application shall support modern mobile browsers.
- The interface shall adapt to different screen sizes.
- The interactive map shall remain usable on touch devices.
- The application shall maintain consistent core functionality across supported devices.

NFR5: Security

The system shall protect the application and administrative functionality from unauthorised access.

- Administrator functions shall require authentication.
- Administrator authorisation shall be enforced server-side.
- Public users shall not be able to modify system data.
- User input shall be validated before processing or storage.
- The system shall protect against common web vulnerabilities.
- Administrator credentials shall be securely stored and transmitted.
- Administrative activities should be logged where appropriate.

NFR6: Privacy

The system shall protect user privacy and minimise the collection of personal information.

- Public users shall not be required to create an account.
- Parking Full reports shall not require user identification.
- The system shall not unnecessarily store the user's precise location.
- Browser geolocation shall only be accessed after permission has been granted.
- The system shall continue functioning when geolocation permission is denied.
- User comments and ratings shall not unnecessarily expose personal information.

NFR7: Reliability

The system shall remain stable when normal operating conditions or external services change.

- The system shall handle unavailable geolocation without crashing.
- The system shall handle invalid destination searches without crashing.
- The system shall handle situations where no parking is available.
- The system shall handle network connectivity loss gracefully.
- The system shall handle external mapping-service failures gracefully.
- Temporary Parking Full reports shall expire automatically.
- The system shall prevent invalid parking areas from being selected.

NFR8: Data Integrity

The system shall maintain accurate and consistent building, parking and activity information.

- Required building and parking information shall be validated before being stored.
- Parking capacity shall contain valid values.
- Parking locations shall contain valid geographic information.
- Parking eligibility shall be consistently applied according to the selected User Category.
- Parking restrictions shall be consistently applied.
- Administrator-approved changes shall be reflected consistently in the public application.
- Temporary Parking Full reports shall not permanently alter administrator-maintained parking data.
- Completed parking activity shall be stored accurately for statistical reporting.

NFR9: Maintainability

The system shall be structured to allow future maintenance and modification.

- The system shall use a modular architecture.
- Map functionality should be separated from parking-management functionality.
- Parking selection logic should be separated from the user interface.
- Public functionality should be separated from administrator functionality.
- Building and parking data should be maintained independently from presentation code.
- The system should allow additional buildings and parking areas to be added without significant modification to the core application.

NFR10: Scalability

The system shall support future expansion of PukParking.

- Additional buildings shall be able to be added without redesigning the application.
- Additional parking areas shall be able to be added without redesigning the core system.
- Additional User Categories should be supportable in future versions.
- The system should accommodate an increasing number of users.
- The statistics functionality should accommodate increasing volumes of parking activity and Parking Full reports.

NFR11: Offline Capability

The offline functionality shall provide a clear and reliable experience during connectivity loss.

- Previously cached application resources shall remain accessible.
- Previously loaded building and parking information shall remain accessible.
- The application shall indicate when the user is viewing cached information.
- The application shall indicate when live information cannot be retrieved.
- New Parking Full reports shall only be submitted when connectivity is available.
- Live routing shall only be generated when the required mapping service is available.

NFR12: External Service Integration

The system shall integrate reliably with required external services.

- The system shall integrate with Mapbox for map display and routing.
- The system shall integrate with browser geolocation functionality where permission is granted.
- The system shall handle temporary unavailability of external services.
- External-service failures shall not cause the entire application to become unusable.
- The system shall provide appropriate feedback when an external service is unavailable.

3.4 Use Case



3.5 Appendix

Issues List

ISSUES LIST		
ISSUE ID	**ISSUE DESCRIPTION**	**STATUS**